

GAYATRI CHOUGULE
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Professional Goal

Aspiring software engineer and machine learning specialist with strong programming, problem-solving, and data analysis skills. Leverage a solid foundation in mathematics (linear algebra, real analysis) and hands-on experience in software development and ML to build scalable, innovative solutions. Seek opportunities in software development, machine learning, or data science to contribute to impactful projects at top-tier tech companies.

Education

University of Maryland, College Park

Bachelor of Science in Computer Science & Mathematics minor

Expected Graduation: May 2025 | GPA: 3.6

Relevant Coursework

- **Programming:** OOP, Web Application Development, Organization of Programming Languages
- **Mathematics:** Discrete Structures, Statistics, Probability, Linear Algebra with CS applications, Calculus I-III, Real Analysis
- **Advanced Topics:** Algorithm Analysis, Database Management, Theory of Computation, Data Science, Advanced Data Structures. Planned: Machine Learning, Artificial Intelligence, Numerical Analysis

Experience

UMD FIRE Research in QML (Jun 2023 - Dec 2023)

(Python, Qiskit, IBM, IonQ, Google Colab)

- Collaborated with a team of four to research Shor's Algorithm (8 weeks) and Quantum Convolutional Neural Networks (QCNN) vs. CNN (14 weeks), focusing on encryption techniques, mathematical foundations, and performance differences.
- Learned quantum computing and QML fundamentals, presented weekly findings, and created research posters for academic conferences.
- Built a responsive website for research dissemination, boosting engagement.
- Validated QCNNs across multiple platforms, including IonQ and IBM Qiskit.

CalQuan Software Solutions, Pune, India (Jul 2023 - Aug 2023; Jul 2024 - Aug 2024; Dec 2024 - Jan 2025) (C#, MsSQL, .NET)

- Collaborated with clients to gather requirements and implemented tailored software solutions.
- Designed and developed components using C# and Vector Draw, including tools for ReBar detailing and parametric drawing.
- Refactored and optimized codebases to enhance performance, maintainability, and readability.
- Conducted thorough testing to ensure software reliability and functionality.
- Identified and documented detailed component requirements to streamline project workflows.
- Developed a C# application to streamline database management by programmatically creating tables, procedures, and functions from predefined scripts.
- Designed and implemented SQL statements to define database structures, procedures, and functions, leveraging advanced MsSQL expertise.

Projects

- **CRUD Application:** Designed a registration system with integrated front-end and back-end components, ensuring seamless user experience and robust data validation.
- **Data Structures Implementation:** Developed and tested data structures (Maps, BSTs, LinkedLists) in Java across multiple projects using Eclipse.
- **Mathematical Modeling:** Applied calculus and linear algebra to enhance predictive analytics accuracy by 10% in data science projects.
- **Python Projects:** Completed 15+ Python projects, including DFA implementations, advanced data structures (KD-trees, splay trees, compressed tries), and data analysis workflows for datasets of 10,000+ entries.

Certifications: Coursera: C Programming, AWS Cloud Management, Crash course in Python

Skills:

- **Programming Languages:** Java, Python, JavaScript, OCaml, Rust, C, C#, SQL, HTML, CSS
- **Mathematical Skills:** Statistical Analysis, Data Modeling, Optimization Techniques
- **Frameworks & Tools:** .NET, DevExpress, Vector Draw, Qiskit, Python Pandas, Google Colab, LINUX CLI, Git, AWS
- **Databases:** SQL, MsSQL
- **Soft Skills:** Problem Solving, Time Management, Communication, Team Collaboration, Attention to Detail, Analytical Thinking